The Last Dance

Time limit per test: 2 seconds
Memory limit per test: 1 gb

Every great story has a final chapter.

In the summer of 2026, an old maestro stepped onto the pitch for one last tournament. The fans had dreamed of a second silverware, but this time it slipped away. The story was already the greatest ever told, yet the fans refuse to let that summer fade.

The archive keeps **N** moments of footage from that summer, numbered from **1** to **N** in the order they happened. The moment numbered **i** carries a tag, an integer a_i written on it by the archivists.

An editor can cut a **reel** from a stretch of the archive. To cut a reel from the stretch [**L**, **R**], the editor picks any non-empty set of moments whose numbers lie between **L** and **R** (inclusive), and the reel shows the picked moments in the order they happened. Two reels are considered different if their sets of picked moments are different, even if they show the same tags.

The fans call a reel a **legend** if it still tells the story of that summer: it must be possible to delete some moments from the reel (possibly none) so that exactly four moments remain, and their tags, in order, are

$$2, 0, 2, 6$$

The archivists will ask **Q** questions. In each question they name a stretch [**L**, **R**] and want to know how many different legend reels can be cut from it.

The answers can be enormous, so print each of them modulo $10^9 + 7$.

Input:

The first line contains a single integer **t** ($1 <= \mathbf{t} <= 10^4$) representing the number of test cases.

Each test case consists of:

1) A line containing two integers **n** and **q** ($1 <= \mathbf{n}, \mathbf{q} <= 2 \times 10^5$) representing the number of moments and the number of questions.

2) A line containing **n** integers $a_1, a_2, ..., a_n$ ($0 <= a_i <= 10^9$) representing the tags of the moments.

3) **q** lines, each containing two integers **L** and **R** ($1 <= \mathbf{L} <= \mathbf{R} <= \mathbf{n}$) describing one question.

It is guaranteed that the sum of n over all test cases does not exceed 2 x 10^5, and the sum of q over all test cases does not exceed 2 x 10^5.

Output:

For each question, print a single integer on its own line: the number of different legend reels that can be cut from the stretch [**L**, **R**], modulo $10^9 + 7$.

Print the answers in the order the questions are given, test case after test case.

Example:

Input
2 6 4 2 2 0 2 6 6 1 5 3 6 2 6 1 6 7 3 20 2 2026 0 26 2 6 1 7 2 7 1 6
Output
3 0 3 9 8 4 0

Explanation:

Test case 1

Question [1, 5]: the stretch shows the tags 2, 2, 0, 2, 6. A legend reel must contain moment 3 (the only 0), moment 4 (the only 2 after it) and moment 5 (the 6), together with at least one of moments 1 and 2. The 3 legend reels are {1, 3, 4, 5}, {2, 3, 4, 5} and {1, 2, 3, 4, 5}.

Question [3, 6]: the stretch shows the tags 0, 2, 6, 6. There is no 2 before the 0, so no reel is a legend.

Question [2, 6]: moments 2, 3 and 4 must be picked, together with at least one of moments 5 and 6, which gives 3 legend reels.

Question [1, 6]: moments 3 and 4 must be picked, together with at least one of moments 1 and 2 (3 ways) and at least one of moments 5 and 6 (3 ways), which gives 3 x 3 = 9 legend reels.

Test case 2

Question [1, 7]: moments 2, 4, 6 and 7 (tags 2, 0, 2, 6) must be picked. Moments 1, 3 and 5 (tags 20, 2026 and 26) can never be among the four remaining moments, so each of them can be picked or not, which gives 2 x 2 x 2 = 8 legend reels. Note that the moment tagged 2026 is a single moment with a single tag.

Question [2, 7]: the same reasoning applies, but only moments 3 and 5 are free, which gives 4 legend reels.

Question [1, 6]: no moment in the stretch is tagged 6, so the answer is 0.

Notes:

Reels are counted by the set of picked moments, not by the tags they show.

Questions do not change the archive; each question is answered independently.